

# A Self-Gated Adaptive Hill- $q$ Rule for Long-Peptide Aggregation

IRIS methodological development memo

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## Abstract

IRIS scores candidate minimal-epitope–HLA pairs, whereas the PDAC and SARS-CoV-2 assays label unresolved long peptides. A Level 2 rule must therefore combine a variable roster of candidate scores into one long-peptide score. Maximum aggregation represents one-candidate sufficiency but ignores corroboration. Log-sum-exp accumulates corroboration but can reward candidate opportunity. The proposed self-gated Hill- $q$  rule lies between these endpoints. It starts at the strongest candidate, forms a score-weighted corroboration offer from the roster, and admits only as much of that offer as the resulting aggregate still needs. The Hill order  $q$  controls how candidate breadth is recognized. The parameters  $(q, c, \kappa)$  may be calibrated for a component model and metric branch, but they remain fixed across the PDAC and SARS-CoV-2 contexts to which that scoring system is applied.

## 1 Why a Level 2 aggregation rule is needed

### 1.1 The experimental unit is a long peptide

For a candidate minimal epitope  $e_i$  and restricting HLA allele  $h_i$ , IRIS produces a positive mechanistic score

$$s_i = F_i(e_i, h_i). \quad (1)$$

The details of  $F_i$  belong to Level 1. For Level 2, the important fact is that a long peptide  $\ell$  can contain a roster

$$\mathcal{C}_\ell = \{(e_i, h_i) : i = 1, \dots, K_\ell\} \quad (2)$$

of several candidate minimal-epitope–HLA pairs.

The observed PDAC and COVID response label belongs to  $\ell$ , not to an identified member of  $\mathcal{C}_\ell$ . Level 2 must therefore map all  $K_\ell$  candidate scores to one long-peptide score  $S_\ell$ .

Because the Level 1 score is positive and multiplicative, it is convenient to work on the log scale:

$$z_i = \log(s_i + \epsilon), \quad (3)$$

where  $\epsilon$  is the declared numerical floor. The transformation is computational; neither  $s_i$  nor  $\text{sigmoid}(z_i)$  is a response probability. The Level 2 map developed below is therefore

$$S_\ell = S(z_1, \dots, z_{K_\ell}). \quad (4)$$

## 1.2 The two biological limits

Two plausible configurations motivate the aggregation problem.

**One-candidate sufficiency.** One candidate may be strong enough to explain the long-peptide response. Other roster members should not dilute it or automatically improve its score.

**Multi-candidate corroboration.** Several candidates may jointly provide persuasive evidence even when no single candidate is decisive. Their magnitudes should be allowed to accumulate.

Maximum aggregation represents the first configuration:

$$S_{\max}(z) = \max_i z_i. \quad (5)$$

It preserves the strongest candidate but discards every other score.

Log-sum-exp represents the second:

$$S_{\text{LSE}}(z) = \log \sum_i e^{z_i}. \quad (6)$$

It accumulates all candidates, but it can increase merely because a peptide offers more candidate opportunities. The adaptive rule is designed to move between these limits according to the score geometry of the individual roster.

## 1.3 What the original PDAC–COVID round robin showed

The motivating comparison crossed five component constructions with twelve fixed Level 2 aggregation rules, producing 60 complete systems. Its results did not identify one aggregation behavior that was adequate in every disease context.

- In the threshold-zero COVID-only comparison, 11 systems were graph-maximal under PR and two under AUROC. The two common systems paired Mono-Q/full-PN with either full log-sum-exp or top-10 log-sum-exp. Within that component family, those two accumulation rules were supported over maximum, mean, median, log-mean-exp, the tested top-fraction means, and the tested top- $k$  means in both COVID datasets and both metrics.
- Original PDAC produced 14 PR-maximal and 13 AUROC-maximal systems, with ten common to both metrics. That set did not contain the COVID Mono-Q/full-PN log-sum-exp system and instead contained several log-mean-exp, top-fraction, top- $k$  mean, and top- $k$  log-sum-exp constructions.
- When PDAC, COVID spike, and COVID non-spike were required jointly, the formal frontier broadened to 44 PR-maximal and 42 AUROC-maximal systems, with 41 in their descriptive overlap. No system was separately maximal in all three contexts under either metric.

**Assay geometry contextualizes the contrast.** The aggregation rules did not face comparable candidate-opportunity sets. COVID spike used 15-mers, and 1,564 of 1,607 COVID non-spike endpoints were also 15-mers. PDAC used predominantly 27-mers: 182 of 213 endpoints had length

27, with a complete range of 15–49 amino acids. If candidate minimal epitopes have lengths 9–12, a peptide of length  $L \geq 12$  contains

$$W(L) = \sum_{r=9}^{12} (L - r + 1) = 4L - 38 \quad (7)$$

sequence windows before HLA expansion and scoreability filtering. Thus a 15-mer offers 22 windows and a 27-mer offers 70. The committed prediction artifacts show that this difference persisted after filtering (Table 1). These are descriptive audits of the original tournament inputs, not additional selection criteria.

Table 1: Long-peptide geometry and realized candidate rosters.  $K$  counts unique scoreable minimal-epitope–HLA tuples. Potential windows refer to the predominant peptide length before HLA expansion and filtering; COVID labels use the threshold-zero definition.

| Cohort          | Length    | Windows | Median (mean) $K$ | $K = 0$ | Mean $K$ , neg./pos. |
|-----------------|-----------|---------|-------------------|---------|----------------------|
| COVID spike     | 15        | 22      | 1 (2.07)          | 36.0%   | 1.30/2.48            |
| COVID non-spike | mostly 15 | 22      | 3 (3.46)          | 32.7%   | 1.32/6.28            |
| PDAC            | mostly 27 | 70      | 10 (10.84)        | 0%      | 10.63/12.57          |

**Sparse COVID rosters made accumulation selective.** Fewer than one percent of the 3,372 COVID endpoints contained more than ten candidates. Full and top-10 log-sum-exp therefore differed on only a small fraction of endpoints, so their joint survival supports soft accumulation over a generally sparse roster rather than a biologically distinguished cutoff at ten. Conversely, because no COVID roster contained more than 15 candidates, the tested 1%, 2%, and 5% top-fraction means each selected one candidate and were exactly score-vector-identical to maximum. Their separate graph vertices do not represent distinct aggregation behavior. Candidate opportunity was also aligned with the threshold-zero labels: positive endpoints carried more candidates than negative endpoints, especially in non-spike, and zero-candidate endpoints were concentrated among negatives. Among nonempty rosters, the mean available LSE bonus  $B$  was 0.517 for non-spike positives and 0.285 for negatives; the corresponding spike values were 0.355 and 0.288. Corroborative mass therefore tended to add more evidence to COVID positives than to negatives.

**Dense PDAC rosters made unrestricted accumulation less selective.** Every PDAC endpoint contained at least one candidate, and 46% contained more than ten. The median available LSE bonus was 0.852, compared with 0.325 in COVID non-spike and 0.230 in COVID spike, but substantial bonus mass was common to both PDAC classes: the mean was 0.910 for positives and 0.791 for negatives. Full log-sum-exp could therefore reward the broad candidate opportunity supplied by a long peptide even when that breadth did not sharply distinguish response. Maximum removed the explicit accumulation bonus but also discarded potentially meaningful second- and third-ranked candidates; it does not by itself eliminate opportunity effects on the extreme score. The PDAC survival of top-fraction, top- $k$  mean, and top- $k$  log-sum-exp rules is consistent with a compromise that retains restricted upper-tail corroboration without accumulating the complete lower-scoring roster.

This interpretation separates Level 2 geometry from Level 1 component choice. The Full-HLA and Mono-Q/full-PN constructions used the same candidate counts; changing the source of  $Q$

changed score magnitudes and ordering, not roster size. Candidate density can therefore help explain why the cohorts favored different aggregation behavior, but it cannot by itself explain the COVID preference for Mono-Q/full-PN at a fixed aggregation rule. Nor do the observed differences establish peptide length as their cause: the cohorts also differ in disease, antigen source, immune history, assay design, HLA distribution, and label construction.

The contrast nevertheless supplies the empirical motivation for an adaptive rule. COVID supports corroborative accumulation where realized breadth is sparse and response-associated, whereas PDAC prevents full log-sum-exp from serving as a universal answer when candidate opportunity is dense and nearly ubiquitous. Rather than choosing one fixed point between maximum and log-sum-exp for every peptide, the proposed rule responds to the realized leader, relative score mass, and candidate breadth of each roster. It also lets admitted corroboration close its own gate as the aggregate becomes stronger. This self-limitation addresses unrestricted accumulation without treating cohort identity or peptide length as an input. It does not, by itself, identify or remove every source of candidate-opportunity confounding. The formal design requirements are stated after the rule is constructed.

## 2 Constructing the adaptive rule

The rule can be summarized in one sentence:

Start with the strongest candidate, form a score-weighted corroboration offer from the roster, and admit only as much of that offer as the resulting aggregate still needs.

### 2.1 Step 1: identify the leader and the available LSE bonus

Let

$$m = \max_i z_i \tag{8}$$

be the strongest candidate score. Express every candidate relative to that leader:

$$r_i = e^{z_i - m}, \quad R = \sum_i r_i. \tag{9}$$

At least one  $r_i$  equals one, and all others lie in  $(0, 1]$ .

The full bonus available between maximum and log-sum-exp is

$$B(z) = \log R. \tag{10}$$

Indeed,

$$S_{\text{LSE}}(z) = m + B(z). \tag{11}$$

Thus  $m$  is the maximum candidate score and  $B$  is the available LSE bonus.

## 2.2 Step 2: measure candidate breadth at a declared Hill order

Normalize the relative strengths:

$$p_i = \frac{r_i}{R}. \quad (12)$$

The  $p_i$  sum to one and describe how the represented score mass is distributed across candidates. They are normalized weights, not response probabilities.

For Hill order  $q \in [0, \infty]$ , define the effective candidate number

$$N_q(p) = \begin{cases} K, & q = 0, \\ \exp\left(-\sum_i p_i \log p_i\right), & q = 1, \\ \left(\sum_i p_i^q\right)^{1/(1-q)}, & q \notin \{0, 1, \infty\}, \\ \frac{1}{\max_i p_i}, & q = \infty. \end{cases} \quad (13)$$

Map it to the bounded breadth coefficient

$$D_q(z) = 1 - \frac{1}{N_q(p)}. \quad (14)$$

For every order,  $D_q = 0$  for a singleton and  $D_q = 1 - 1/K$  when the  $K$  candidates have equal weights. The orders differ only for uneven rosters. Hill numbers are nonincreasing in  $q$ : low orders recognize more of the roster as breadth, whereas high orders require corroborative mass to lie closer to the leader.

| Order        | Breadth interpretation                                                                      |
|--------------|---------------------------------------------------------------------------------------------|
| $q = 0$      | Richness-sensitive: every unique candidate contributes to $D_q$ .                           |
| $q = 1$      | Entropy-weighted: low-weight candidates contribute in proportion to their information mass. |
| $q = 2$      | Concentration-sensitive: $D_2 = 1 - \sum_i p_i^2$ .                                         |
| $q = \infty$ | Leader-share-sensitive: $D_\infty = 1 - \max_i p_i$ .                                       |

At  $q = 0$ , the breadth coefficient depends on  $K$ , but the complete corroborative quantity below does not depend on count alone: the available bonus  $B$  still weights candidates by their realized scores.

## 2.3 Step 3: form the corroboration offer

Define

$$C_q(z) = B(z)D_q(z). \quad (15)$$

The available bonus  $B$  asks how much relative score mass the roster contains. The coefficient  $D_q$  asks how that mass should count as breadth at the declared Hill order. Their product  $C_q$  is the corroboration offered before sufficiency gating.

## 2.4 Step 4: define an aggregate-sufficiency gate

For any provisional aggregate score  $a$ , define the decreasing gate

$$g_{c,\kappa}(a) = \text{sigmoid}\left(\frac{c-a}{\kappa}\right) = \frac{1}{1 + \exp[(a-c)/\kappa]}, \quad \kappa > 0. \quad (16)$$

The center  $c$  is the aggregate score at which the gate equals one-half. The width  $\kappa$  controls how gradual the transition is: the gate falls from 0.9 to 0.1 over an interval of length  $2 \log(9)\kappa \approx 4.39\kappa$ .

A low aggregate leaves the gate open and permits corroboration. As admitted corroboration raises the aggregate, the same gate progressively closes. A strong leader starts the aggregate high and therefore makes the rule max-like from the outset.

## 2.5 Step 5: let the aggregate gate itself

The self-gated adaptive Hill- $q$  aggregate is the unique solution of

$$S_\ell = \underbrace{m}_{\text{maximum candidate score}} + \underbrace{C_q(z)}_{\text{corroboration offer}} \times \underbrace{g_{c,\kappa}(S_\ell)}_{\text{aggregate-sufficiency fraction}}. \quad (17)$$

The equation is implicit by design. The gate is evaluated at the result after corroboration has been admitted, rather than at the pre-aggregation maximum candidate score  $m$  before admission. The rule therefore admits corroboration only while the resulting aggregate still leaves room for it.

| Quantity               | Interpretation                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $m$                    | <b>Maximum candidate score.</b> This is the score of the single best candidate and therefore the evidence already available without invoking any corroboration. Starting from $m$ guarantees that additional candidates can only add to, rather than dilute, the strongest signal; for a singleton, the aggregate reduces exactly to $m$ .                                                      |
| $B$                    | <b>Available LSE bonus.</b> Because $B = S_{\text{LSE}} - m = \log \sum_i e^{z_i - m}$ , it is the entire gap between max pooling and log-sum-exp pooling. Candidates near the leader contribute substantially to this offer, whereas candidates far below the leader contribute little. Crucially, $B$ is only the amount <i>available</i> ; the rule does not accept all of it automatically. |
| $D_q$                  | <b>Breadth fraction at Hill order <math>q</math>.</b> The order determines how strongly low-weight candidates count as breadth. At $q = 0$ , every unique candidate contributes to the coefficient; at larger $q$ , consequential mass must be increasingly concentrated near the leader.                                                                                                       |
| $C_q$                  | <b>Corroboration offer.</b> The product $BD_q$ combines realized relative score mass with the declared interpretation of breadth. It is the largest bonus available before the aggregate-sufficiency gate is applied.                                                                                                                                                                           |
| $g_{c,\kappa}(S_\ell)$ | <b>Fraction permitted by aggregate sufficiency.</b> This fraction is near one while the developing aggregate is low, equals one-half at $S_\ell = c$ , and approaches zero as $S_\ell$ becomes high. A large offer therefore helps close its own gate rather than being accepted without limit.                                                                                                 |

## 2.6 Why the implicit definition is safe

Let  $C = C_q(z)$  and define

$$H(a) = a - m - Cg_{c,\kappa}(a). \quad (18)$$

If  $C = 0$ , Equation (17) gives  $S_\ell = m$ . If  $C > 0$ , then

$$H(m) = -Cg_{c,\kappa}(m) < 0, \quad H(m + C) = C\{1 - g_{c,\kappa}(m + C)\} > 0. \quad (19)$$

Moreover, because the gate is strictly decreasing,

$$H'(a) = 1 - Cg'_{c,\kappa}(a) > 0. \quad (20)$$

There is therefore exactly one solution in  $[m, m + C]$ . It follows immediately that

$$m \leq S_\ell \leq m + C_q(z) \leq m + B(z) = S_{\text{LSE}}(z). \quad (21)$$

The score always lies between maximum and log-sum-exp. Increasing the corroboration offer cannot lower the solution: implicit differentiation gives

$$\frac{\partial S_\ell}{\partial C} = \frac{g_{c,\kappa}(S_\ell)}{1 - Cg'_{c,\kappa}(S_\ell)} > 0. \quad (22)$$

This monotonicity is one reason to prefer the self-consistent gate over the explicit shortcut  $m + Cg_{c,\kappa}(m + C)$ , whose accepted bonus can decrease when the offer becomes larger.

The unique root can be computed deterministically by bisection on  $[m, m + C]$ . The interval is known in advance, the root is scalar, and no naive fixed-point iteration is required. Solver tolerance and iteration limits form part of the scoring specification.

## 2.7 Full design requirements and roster comparability

The construction separates three questions. The available bonus  $B$  asks how much relative score mass is present. The Hill coefficient  $D_q$  asks how the distribution of that mass should count as breadth. The self-gate  $g(S_\ell)$  asks how much corroboration the resulting aggregate still needs. A complete adaptive Level 2 design must also state how it treats differences in roster construction without assuming that roster size is either automatically supportive or automatically suspect.

It is useful to state the full design problem explicitly. An adaptive long-peptide aggregate should solve, or at least expose, each of the following problems.

- P1. Unresolved causation.** The response belongs to a long peptide, but the causal minimal epitope–HLA pair is not observed. The rule must aggregate the candidate roster without pretending that it has identified which candidate caused the response.
- P2. One-candidate sufficiency.** One strong candidate may be sufficient. Additional weak candidates should neither dilute that leader nor automatically create appreciable evidence.
- P3. Meaningful corroboration.** Several consequential candidates may jointly support a response even when no single candidate is decisive. A rule that always returns the maximum discards this information.

- P4. Dominance versus breadth.** A roster containing one leader and many negligible candidates should be distinguished from a roster in which several candidates carry comparable score mass. The available bonus  $B$  records total relative mass, while  $D_q$  controls how its distribution is interpreted as breadth. At low  $q$ , more of the roster contributes; at high  $q$ , breadth must be concentrated nearer the leader. The rule should make this choice explicit rather than embedding one privileged order without examination.
- P5. Aggregate sufficiency.** The same corroboration offer need not have the same meaning at every absolute score level. A strong leader may need little help. A weak leader may admit corroboration, but the admitted amount should diminish as the resulting aggregate itself becomes strong. Gating only at the initial leader cannot provide this feedback.
- P6. Candidate-opportunity comparability.** Candidate rosters may differ in size because of peptide length, HLA coverage, or roster-construction choices. The role of roster richness must therefore be declared. At  $q = 0$ , every unique tuple contributes to the breadth coefficient; at higher orders, low-weight tuples contribute less. In every case,  $B$  still controls their realized score mass and the self-gate limits how far the offer moves the aggregate. The rule does not apply an additional penalty based only on  $K$ , and it does not claim to eliminate every opportunity effect caused by peptide length or enumeration depth.
- P7. Exact-duplicate control.** The aggregation operates on unique minimal-epitope–HLA tuples. If the same tuple appears more than once—for example, because a patient is homozygous for the HLA allele—it must enter the roster only once; otherwise repeated representation would manufacture breadth. By contrast, the same minimal epitope paired with different HLA alleles forms distinct tuples and remains as distinct candidates. The adaptive aggregation does not collapse or otherwise specially adjust those cross-HLA candidates.
- P8. Symmetry and singleton coherence.** Candidate order must not matter, tied candidates must be treated symmetrically, and a roster containing one candidate must return that candidate’s Level 1 log score.
- P9. Auditability and label blindness.** Every adaptation term must be calculable from the candidate scores under the prespecified roster convention. Endpoint response labels must not choose the mixing behavior used for that endpoint.

Problems P4 and P6 are especially easy to conflate. Hill breadth describes how a declared order reads one realized weight distribution. Candidate- opportunity comparability asks whether a difference in roster construction represents additional biological evidence, harmless enumeration, or duplicated evidence. The aggregation function exposes the first choice. It cannot settle the second question without assumptions about how candidate opportunities are generated.

The recommended rule therefore does *not* contain a separate roster-size penalty that lowers the aggregate as roster size increases. Such a penalty would assign a lower score to the same meaningful score geometry merely because more candidates were enumerated. That ordering would be defensible only under an explicit stochastic model in which additional candidate opportunities generate spurious favorable scores. Without such a model, the penalty could remove genuine biological signal and make the result depend on pipeline exhaustiveness.

Candidate opportunity enters through both the realized score mass and the declared Hill order. If an additional candidate lies  $d$  log-score units below the leader, its relative weight is  $e^{-d}$ . Given the

previous relative-weight sum  $R$ , its increment to the available LSE bonus is

$$\log(R + e^{-d}) - \log R = \log\left(1 + \frac{e^{-d}}{R}\right) \leq \frac{e^{-d}}{R}. \quad (23)$$

Thus additional candidates have little effect through  $B$  when their *combined* relative score mass is negligible. At  $q = 0$ , however, adding a new unique tuple also changes  $D_0 = 1 - 1/K$ ; this richness sensitivity is an explicit modeling choice rather than an accidental invariance claim. Exact-duplicate control occurs before the score vector  $z$  is formed: the roster is a set, rather than a multiset, of minimal-epitope–HLA tuples. A homozygous recurrence of the same tuple therefore contributes one candidate, whereas the same minimal epitope paired with two different HLA alleles contributes two distinct corroborative contributions. Overlapping but nonidentical minimal epitopes likewise remain distinct. The formula does not assert that these contributions are statistically independent.

The recommended aggregate consequently remains  $S_\ell$ , with no separate roster-size penalty and no cohort-identity input. The self-gate limits accumulation according to the score the roster actually reaches; it should not be described as a complete correction for roster opportunity.

### 3 How the rule behaves

#### 3.1 Limiting cases

Equation (21) guarantees that the result always lies between maximum and log-sum-exp. The fixed point adds a second form of adaptation: a large corroboration offer raises the aggregate, and that increase progressively closes the same gate through which the offer is admitted.

| Candidate pattern         | Offer and gate                     | Result                           |
|---------------------------|------------------------------------|----------------------------------|
| One candidate             | $C_q = 0$                          | exactly maximum                  |
| One dominant candidate    | small $B$ , hence small $C_q$      | approximately maximum            |
| Weak sparse corroboration | modest offer; gate initially open  | substantial partial accumulation |
| Weak dense corroboration  | large offer closes its own gate    | self-limited accumulation        |
| Strong leader             | aggregate starts high; gate closed | max-like behavior                |

Candidate order does not matter. A singleton returns its Level 1 log score exactly. Tied candidates are treated symmetrically. For an empty roster, the declared floor score is returned before Equation (17) is evaluated.

When  $\kappa$  approaches zero, the transition becomes nearly hard. Its behavior is then approximately

$$S_\ell \approx \begin{cases} m, & m \geq c, \\ m + C_q, & m + C_q \leq c, \\ c, & m < c < m + C_q. \end{cases} \quad (24)$$

Thus a strong leader receives essentially no bonus, a weak roster whose full offer remains below  $c$  can receive nearly all of it, and an offer that would cross  $c$  is admitted only until the developing aggregate reaches the gate's transition region. For finite  $\kappa$ , these cases join smoothly rather than forming a strict cap.

### 3.2 Worked score geometries

The following examples use  $q = 2$ ,  $c = -1$ , and  $\kappa = 0.5$  only to illustrate the calculation. Equal-weight plateaus have the same Hill number at every order, so changing  $q$  does not alter those rows.

**Four illustrative candidate-score geometries.**

| Roster             | $z$              | $D_2$ | $g(S_\ell)$ | Max | LSE    | Self-gated result                                                       |
|--------------------|------------------|-------|-------------|-----|--------|-------------------------------------------------------------------------|
| Isolated leader    | $(0, -5, -5)$    | 0.026 | 0.119       | 0   | 0.013  | 0.00004: negligible corroboration                                       |
| Weak plateau       | $(-2, -2, -2)$   | 0.667 | 0.720       | -2  | -0.901 | -1.473: broad evidence receives a substantial bonus                     |
| Strong plateau     | $(0, 0, 0)$      | 0.667 | 0.104       | 0   | 1.099  | 0.076: the strong aggregate closes most accumulation                    |
| Dense weak plateau | ten scores at -2 | 0.900 | 0.491       | -2  | 0.303  | -0.982: the large offer raises the score toward $c$ and closes its gate |

The two three-candidate plateaus have identical breadth and offers but different absolute score levels. The dense weak plateau has a much larger offer than the weak three-candidate plateau. Its aggregate nevertheless stops near the gate center because admitted corroboration progressively reduces  $g(S_\ell)$ . The calculation is endpoint-adaptive: no cohort label or peptide length enters the equation.

## 4 Parameter selection and evidential status

The functional form is shared, but its parameters need not be identical for Level 1 score constructions with different candidate-score geometries. Let  $u$  index the component model and  $b \in \{\text{PR}, \text{ROC}\}$  index the metric branch. The recommended contract selects one triple

$$(q_{ub}, c_{ub}, \kappa_{ub}) \quad (25)$$

for each component-metric group. That triple is then fixed across original PDAC, threshold-zero COVID spike, and COVID non-spike. The aggregation rule therefore remains unaware of dataset identity: context adaptation occurs only through the endpoint's  $m$ ,  $B$ ,  $D_q$ , and self-consistent aggregate.

Selection should evaluate one frozen joint Cartesian grid

$$\mathcal{G} = \mathcal{Q} \times \mathcal{C} \times \mathcal{K} \quad (26)$$

for every component-metric group. Within each cohort, all triples are ranked jointly by the branch metric. Cohorts receive equal weight. The primary choice minimizes the equal-cohort mean fractional rank; declared tie-breaks minimize the worst-cohort fractional rank and then mean metric regret before applying deterministic numerical tie-breaks. PR selection uses ordinary empirical average precision with threshold-block tie handling. ROC selection uses AUROC with half credit for tied positive-negative pairs.

The entire joint  $(q, c, \kappa)$  grid  $\mathcal{G}$  must be frozen before the surfaces are examined. Adding intermediate  $q$ ,  $c$ , or  $\kappa$  values after viewing a rank surface changes a rank-based objective through joint-grid

density. The output should retain the complete surfaces, the selected triple, the number and composition of near-optimal triples, and whether  $c$  or  $\kappa$  lies on a numerical boundary. An endpoint Hill order  $q = 0$  or  $q = \infty$  is a scientifically interpretable member of the frozen candidate set, not an instruction to extend the order grid.

Two diagnostics are especially important. First, report which Hill orders occur among triples within a declared tolerance of the optimum, rather than only the single lexicographic winner. Second, repeat the selection after leaving out each cohort in turn. These analyses distinguish a stable component–metric calibration from a triple whose apparent adequacy depends on one context or a narrow  $\kappa$  transition.

All endpoint adaptation is label blind once the triple is fixed. Response labels participate only in the model-development selection of  $(q, c, \kappa)$ ; they never choose the gate or Hill order for an individual endpoint. If the same PDAC and COVID cohorts later supply a directed round-robin comparison of the selected scores, that evidence is conditional post-selection evidence rather than independent confirmation. The parameter selection, complete surfaces, solver contract, source checksums, and this evidential status should be recorded in the selection manifest.

## 5 Interpretive qualifications

**Gate threshold.** All candidates are represented on the shared Level 1 scale  $z_i = \log(F_i + \epsilon)$ . Accordingly,  $c$  is the aggregate log- $F$  score at which the aggregate-sufficiency fraction equals one-half. It is no longer a threshold applied only to the maximum candidate. The definition of  $F$ , the logarithmic transformation, and  $\epsilon$  must therefore remain fixed when  $c$  and  $\kappa$  are specified.

**Hill order.** The order  $q$  controls how the normalized candidate-weight distribution is translated into breadth. It is a calibration of the Level 2 rule, not a probability of candidate independence. A model–metric-specific  $q$  remains fixed across cohorts after selection.

**Candidate-opportunity comparability.** The rule contains no separate penalty based on roster size. Roster richness can nevertheless enter through  $D_q$ , especially at  $q = 0$ , while  $B$  records the candidates’ combined relative score mass and the self-gate limits accepted accumulation. This is a declared treatment of candidate opportunity, not proof that peptide length or enumeration depth is harmless. Exact duplicate minimal-epitope–HLA tuples are removed before aggregation, including repeated occurrences caused by HLA homozygosity. Distinct overlapping epitopes and the same epitope associated with different HLA alleles remain separate corroborative contributions.

**Numerical solution.** Equation (17) is solved as a scalar root on the certified interval  $[m, m + C_q]$ . The bisection tolerance, maximum iteration count, empty-roster floor, and behavior on any numerical failure must be fixed and audited. A naive fixed-point iteration is not part of the definition.

**Scope of the revision.** The self-gate limits an offered bonus according to the aggregate it produces. It does not establish that the roster convention is biologically complete, that distinct tuples are statistically independent, or that the selected parameters will transfer to an unrepresented context. Those claims require additional evidence.